DO NOW — RECALL

Lesson 108 · Interrupts and the Interrupt Service Routin

Name: _____ Class: _____ Date: _____

1. With page size 1,024 bytes, what offset is address 2,500?

2. For the binary value 1010, what is its decimal value?

3. A linked-list node has data and what other essential field?

4. In a BST, where is a value less than the current node placed?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

452.

2. Model answer

10.

3. Model answer

A link/reference to the next node (for a singly linked list).

4. Model answer

In its left subtree (under the chosen no-duplicate convention).

By the end, you can…

• A device or timer signals that attention is needed.

• The CPU completes a suitable instruction boundary and saves essential context.

• The ISR handles the event; saved context is restored and execution resumes.

Vocabulary — know these terms

interrupt — Signal requiring CPU attention.

ISR — Routine that services a particular interrupt.

context — CPU state needed to resume execution.

polling — Repeated checking of a device or condition.

NEW LEARNING

Read, annotate and explain

Event occurs

A device or timer signals that attention is needed.

State is preserved

The CPU completes a suitable instruction boundary and saves essential context.

Service and return

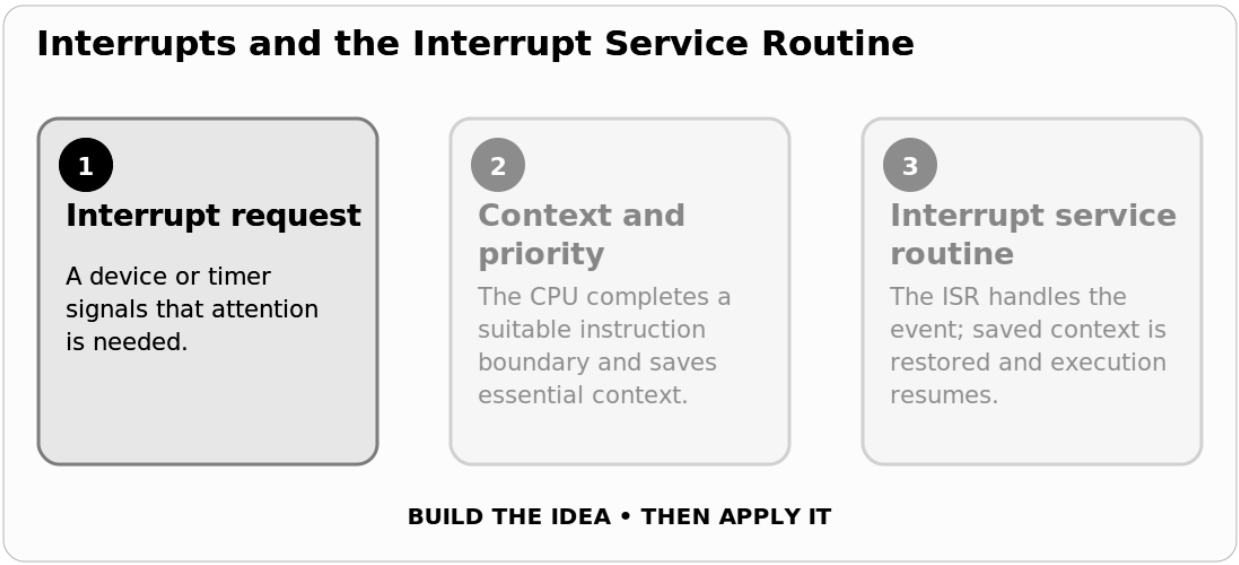
The ISR handles the event; saved context is restored and execution resumes.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

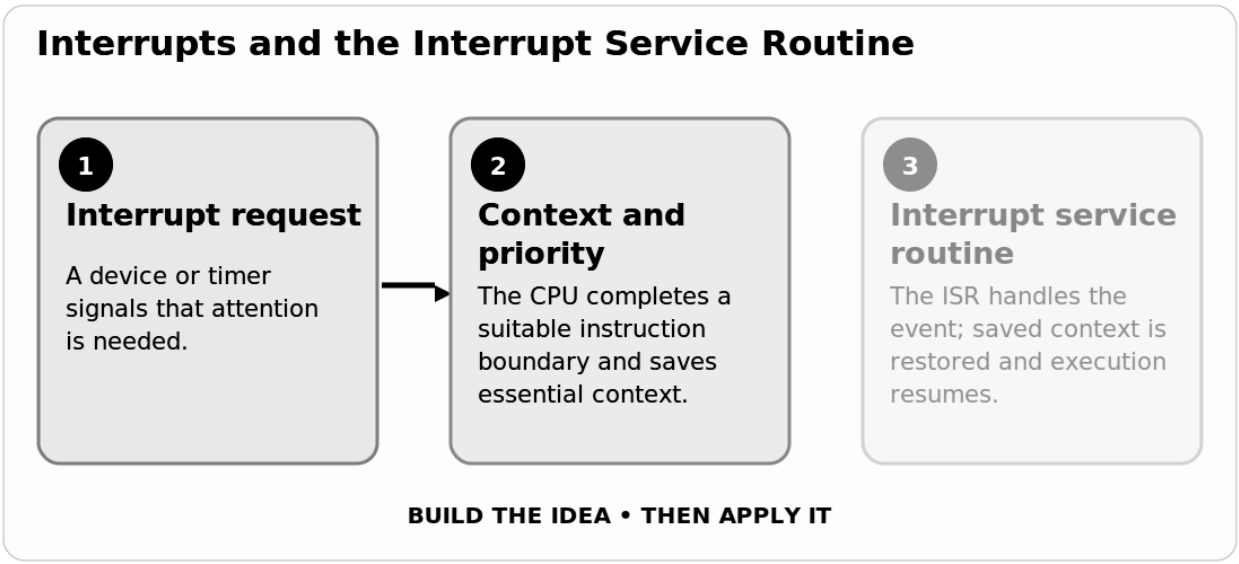
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Interrupts and the Interrupt Service Routine: Interrupt request then Context and priority then Interrupt service routine

New Learning

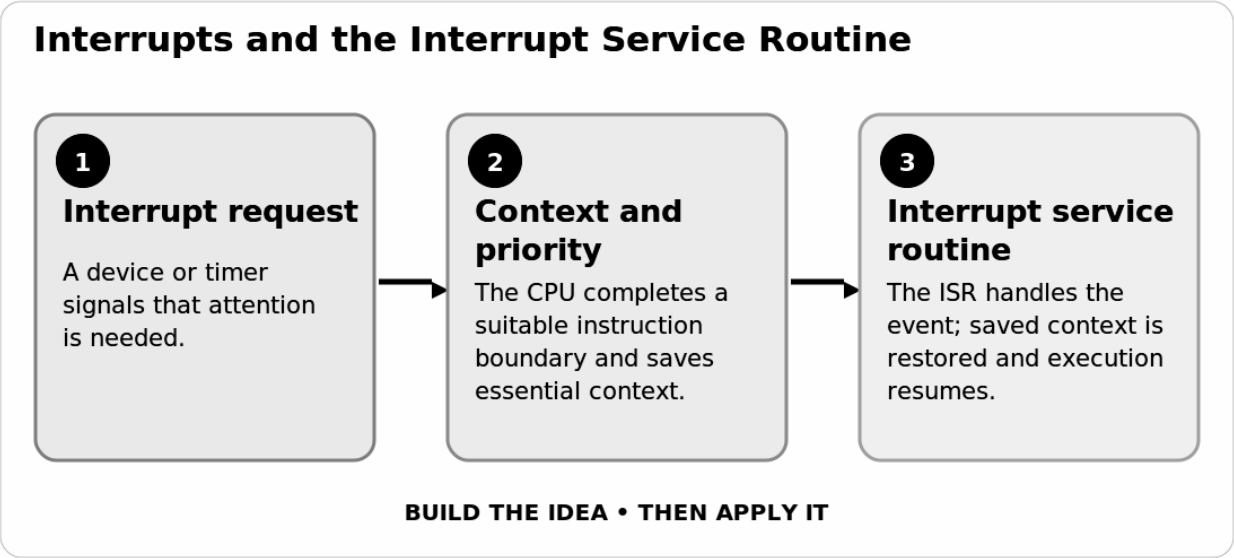


Interrupts and the Interrupt Service Routine: Interrupt request then Context and priority then Interrupt service routine

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Interrupts and the Interrupt Service Routine: Interrupt request then Context and priority then Interrupt service routine

I DO — WORKED EXAMPLE

Follow the worked method

A keyboard interrupt arrives while a calculation is running. Trace the CPU response.

Model answer

After completing the current instruction and recognising the interrupt, the CPU preserves relevant state, runs the keyboard ISR to obtain the key event, restores state and resumes the interrupted program.

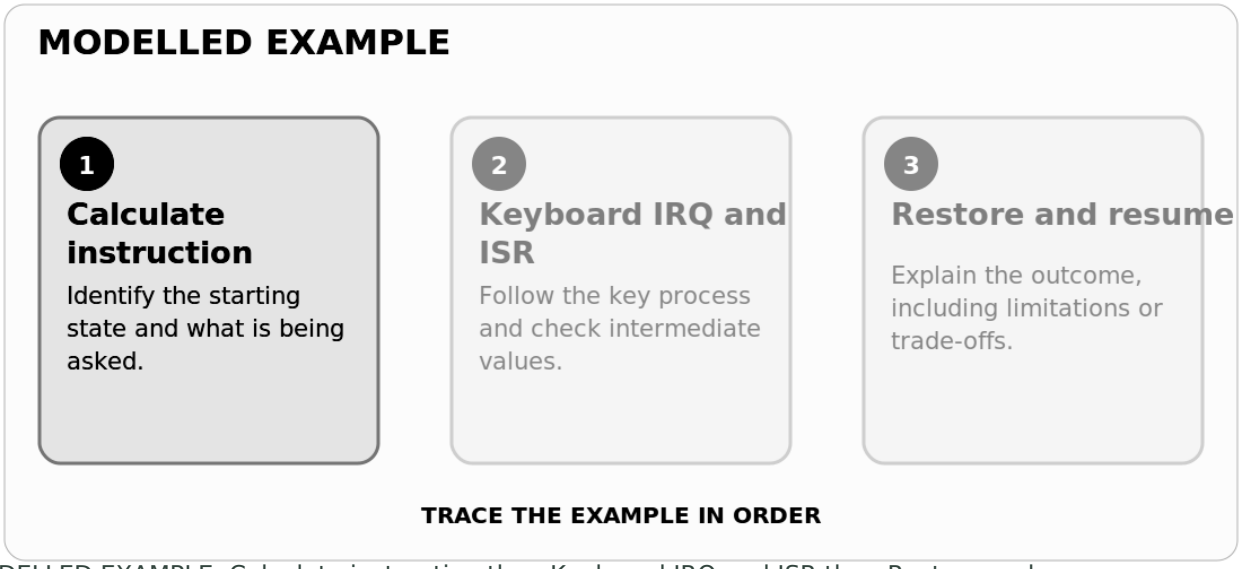
Check: link each claim to the scenario and use precise OCR terminology.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

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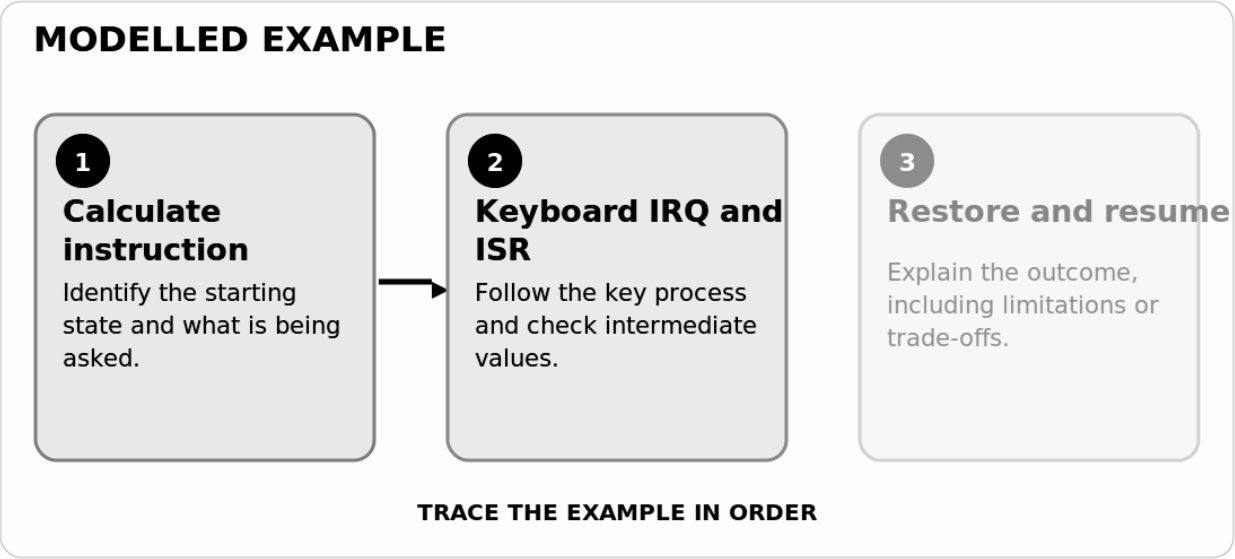
MODELLED EXAMPLE: Calculate instruction then Keyboard IRQ and ISR then Restore and resume

I Do · original lesson diagrams

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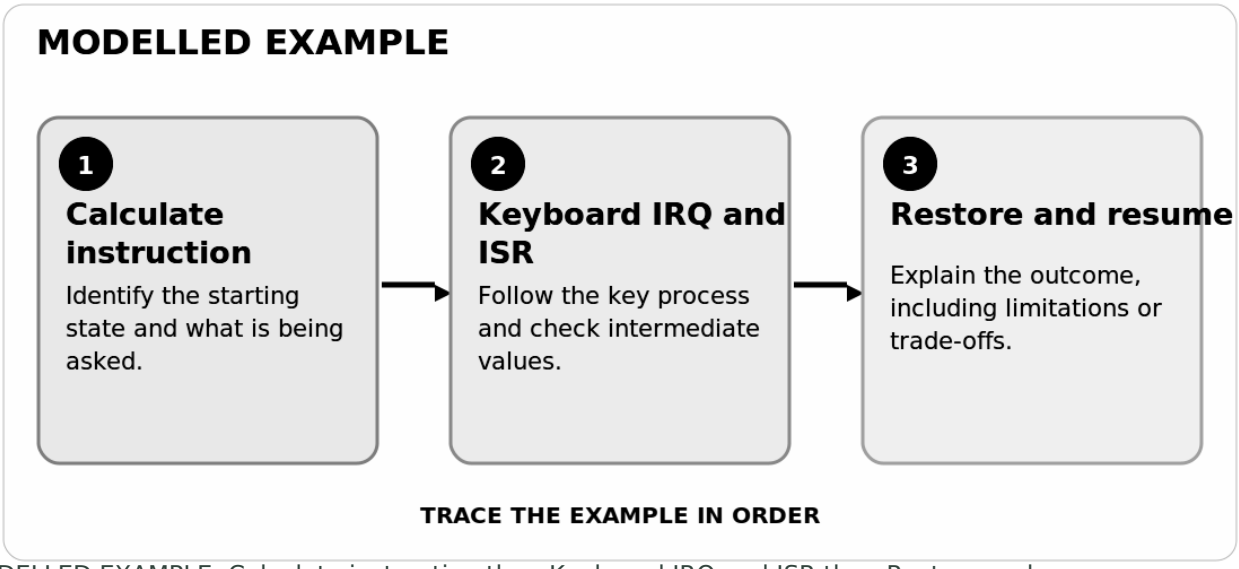
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WE DO — GUIDED PRACTICE

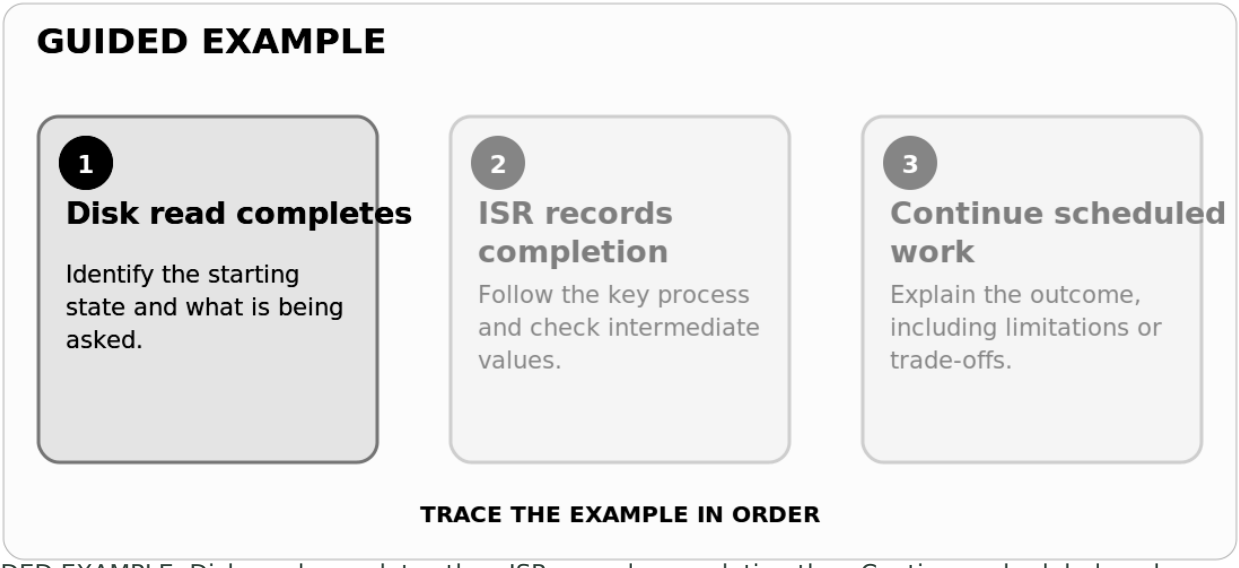
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A disk controller signals that reading a sector has finished. Explain the sequence.

Identify the inputs, relevant facts and exact output or judgement required.



GUIDED EXAMPLE: Disk read completes then ISR records completion then Continue scheduled work

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: What must we decide? Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Outline the role of an interrupt and an ISR. [6]

Question 2

Explain where an interrupt can fit into the fetch–decode–execute cycle. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate interrupts compared with repeatedly polling a low-frequency sensor. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An ISR should normally preserve enough execution state to allow the interrupted work to resume.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

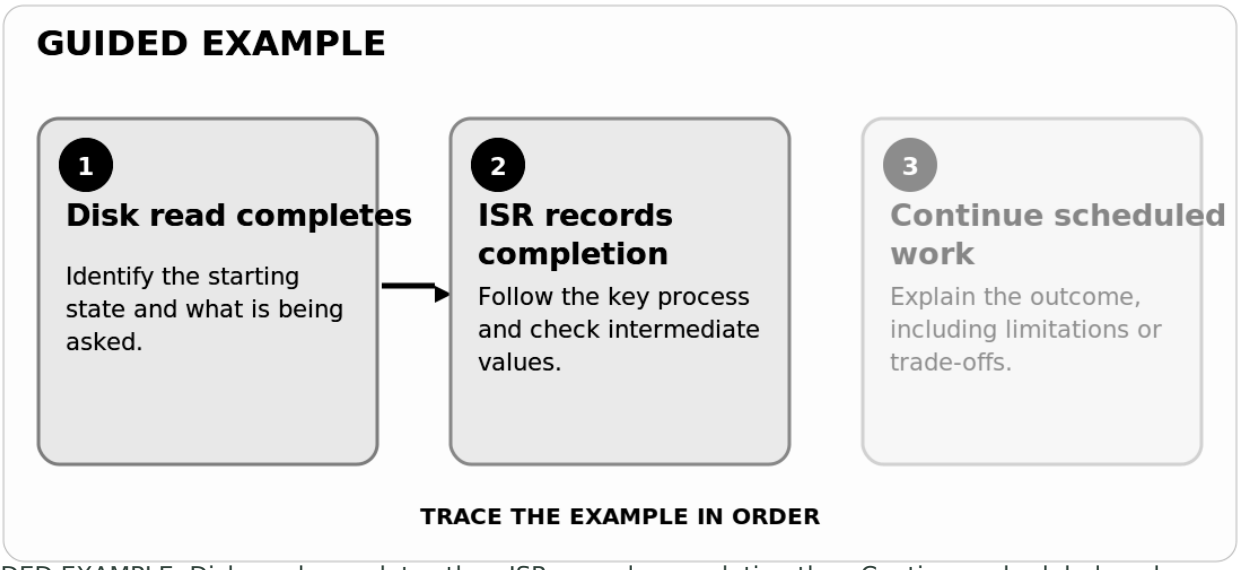
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

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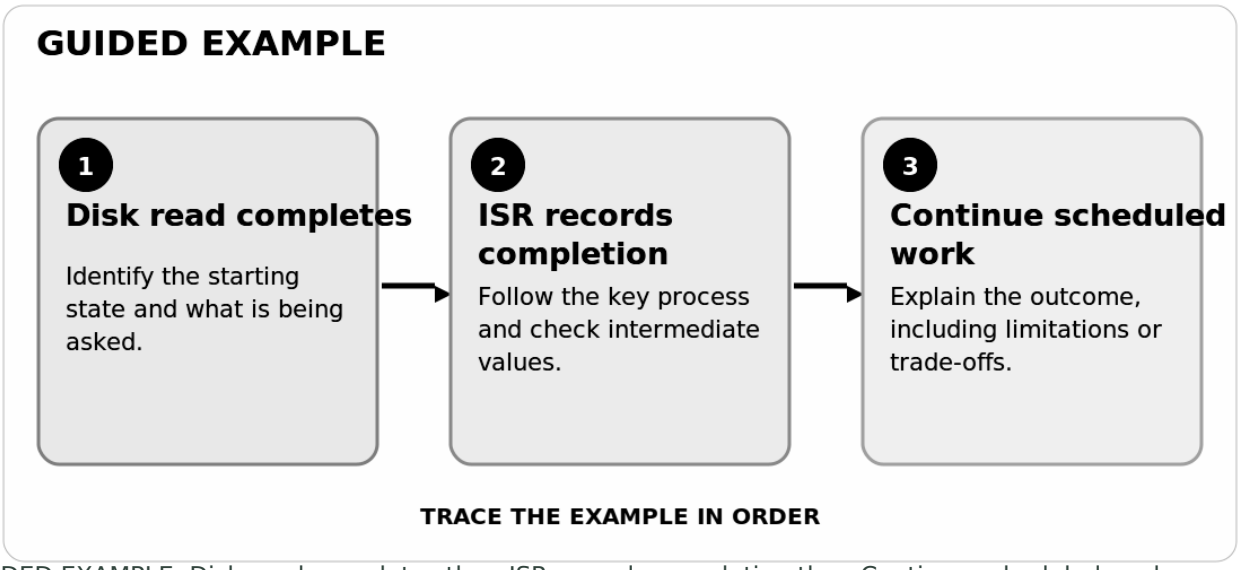
Identify the inputs, relevant facts and exact output or judgement required.



GUIDED EXAMPLE: Disk read completes then ISR records completion then Continue scheduled work

A disk controller signals that reading a sector has finished. Explain the sequence.

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GUIDED EXAMPLE: Disk read completes then ISR records completion then Continue scheduled work

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

The disk raises an interrupt; the CPU saves context when accepted; the ISR acknowledges completion and updates status; the original process resumes or the scheduler selects another ready process.

IMPROVE: Explain one correction to your own reasoning.